How does the model behave

Phase I: black-box behaviour

We only look at input-outputs.

Model & Data

- DieT-Tiny (Vision Trasformer), 5 classes
- Classes : call, dislike, fist, four, like
- HaGRID hand-gesture images, 1000 test images

Confidence – Definition

- Model’s own probablity for it’s top guess.
  - Example : “fist” – 0.99 means 99% sure

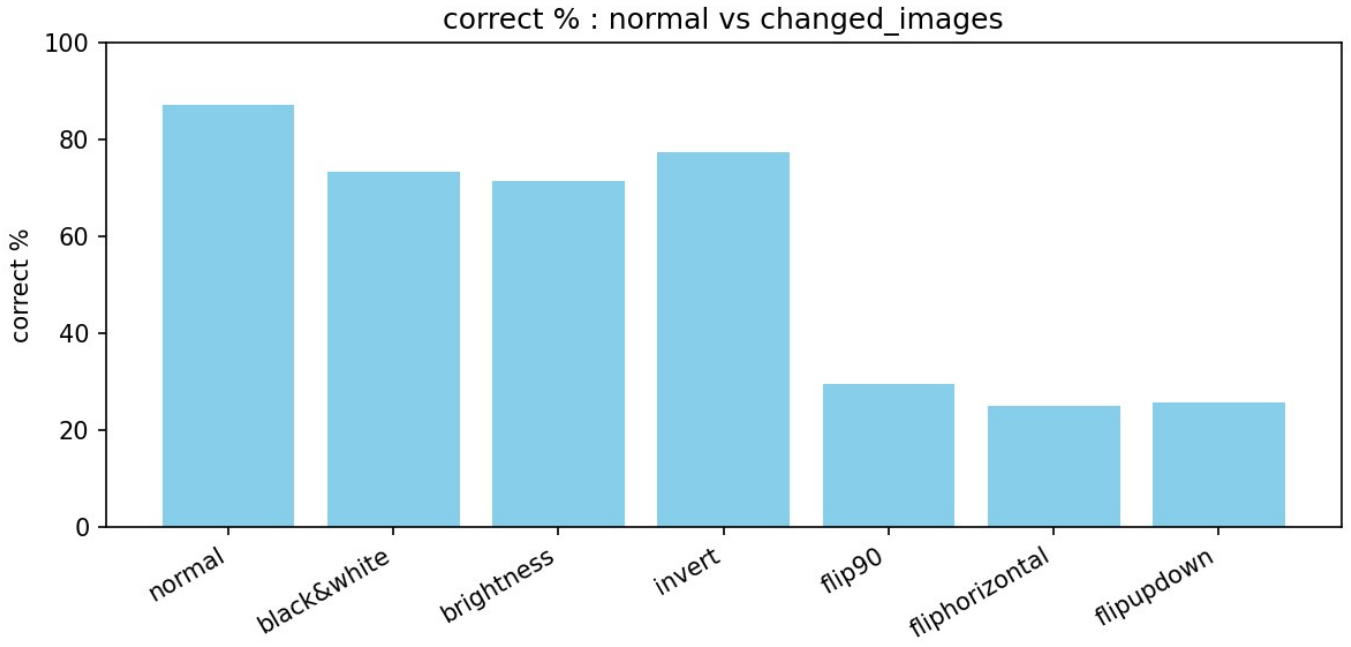
What is trust?

- Can confidence tell us when the model is wrong
  - Ie, if it says 99%, can we trust it

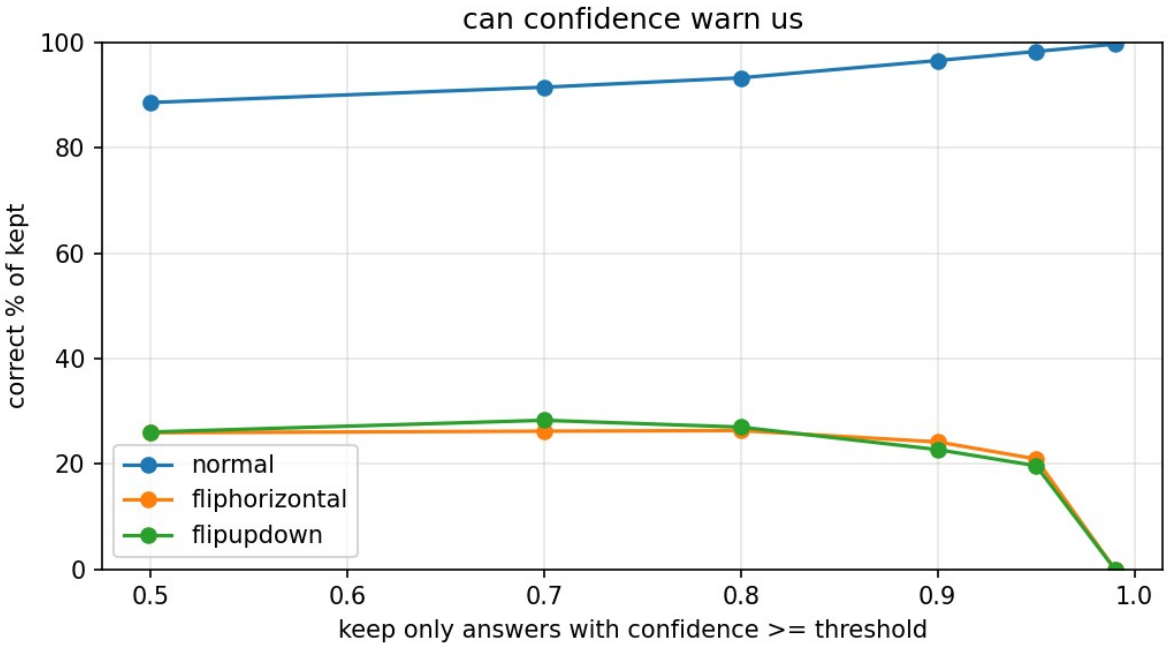
Method

- Keep only answers with confidence >= a cut-off
- Measure : how many we keep, and how many of those are right
- Test cut-offs from 0.50 up to 0.99

Result : accuracy collapses(fig1)

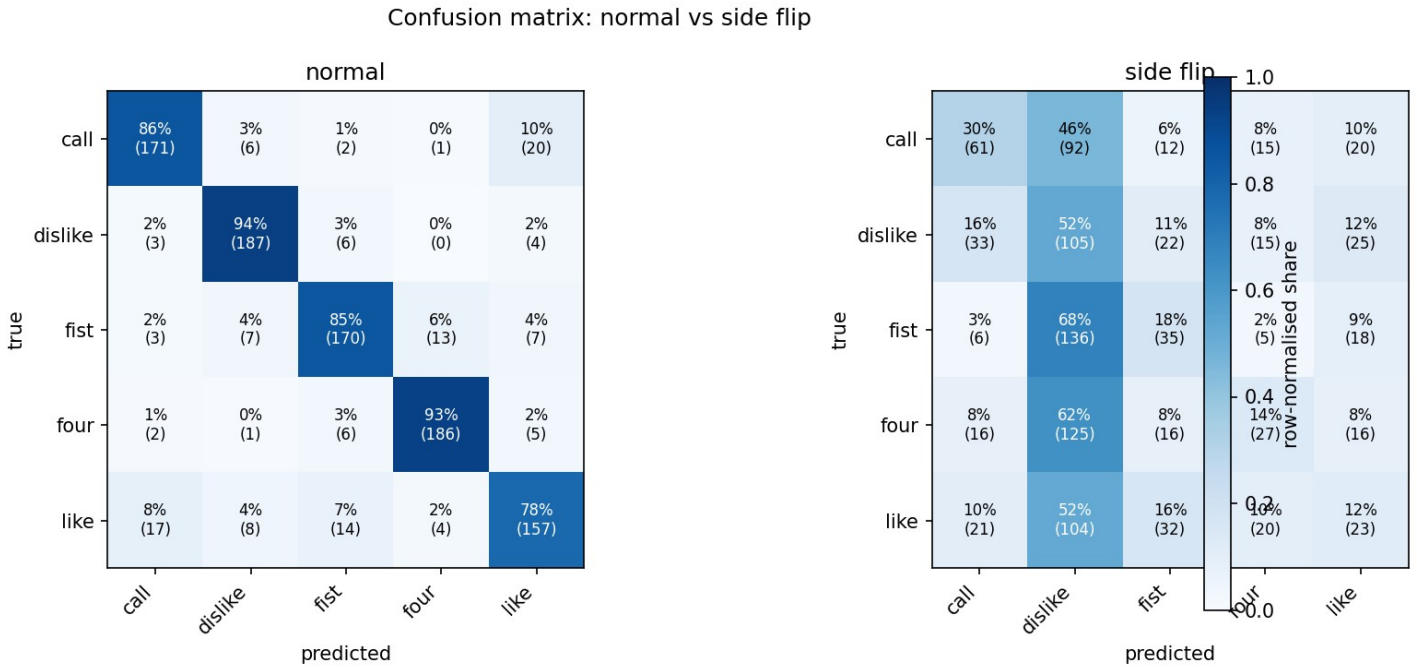


Result : confidence doesnt warn us



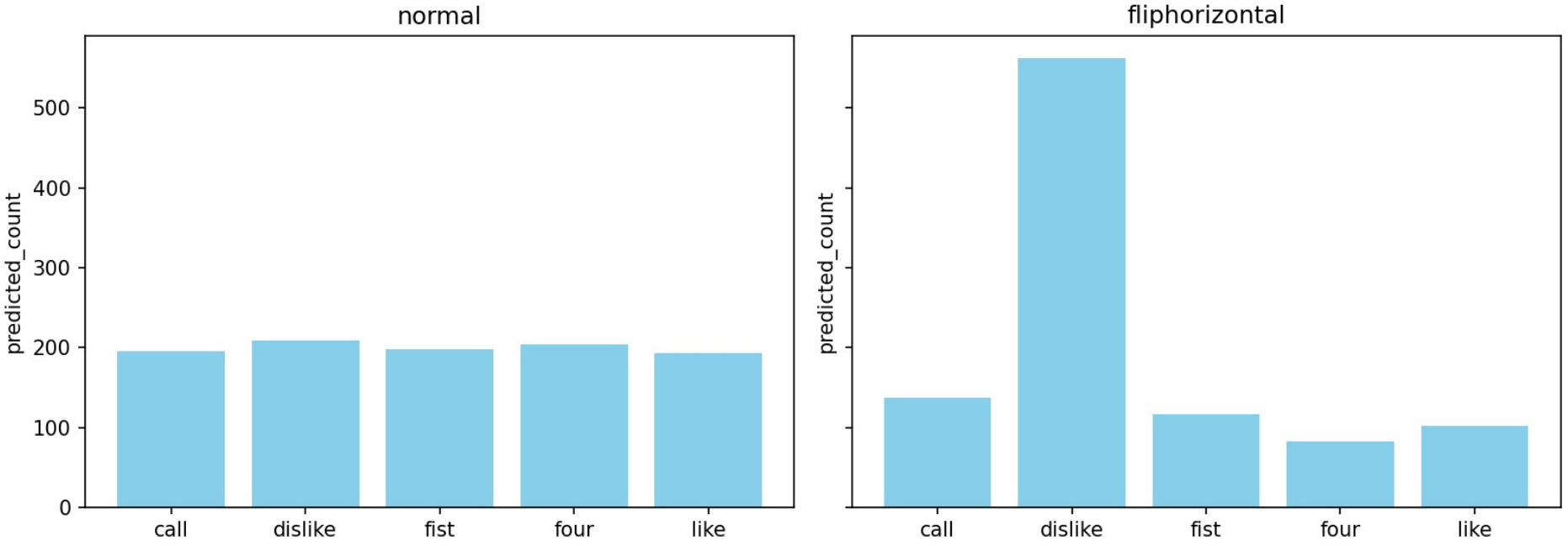
Trust curve: Normal stays high; flipped images stay low

Predictions collapse



After a side flip, most predictions collapse to ‘dislike

Result : Almost Always dislike



8 Numbers

- Normal : 87.1%
- Side flip : 25.1 %
- Sure >= 90% normal : keep 73%, 96.6% right
- Sure >= 90% side flip: keep 24%, 24.2% right

Conclusion

- Confidence is not a safe warning signal when images change.
- Flipped images drop from ~87% to ~25% correct
- Limitaion : Which pixel cause it is still open(Phase II)

Back up : confidence overlap

